

1 Report:Exploratory Data Analysis

NGAsub Subduction Dataset

This report documents a complete exploratory data analysis of the NGAsub flatfile used for ANN and other ML modeling. The EDA focuses on:

- Data coverage and variable ranges.
- Distributional differences between interface and intraslab events.
- Regional balance in terms of records, events, and stations.
- Behavior of key predictors (M_w , R_{jb} , Z_{tor} , and V_{S30}).
- Residual variability trends needed for uncertainty interpretation.

2 Dataset and Variables

The base dataset is `NGAsub_flatfile_filtered.csv`. The analysis partitions records into:

- **Interface events:** `Intra_Inter_Flag = 0`.
- **Intraslab events:** `Intra_Inter_Flag` in $\{1, 5\}$.

The principal variables explored are:

- **Earthquake_Magnitude** (M_w): event magnitude.
- **Rjb_km** (R_{jb}): Joyner–Boore distance in km.
- **Ztor_km** (Z_{tor}): depth to the top of rupture in km.
- **Vs30_Selected_for_Analysis_m_s** (V_{S30}): proxy for site stiffness.
- **DatabaseRegion**: geographic source region.

3 Preprocessing Protocol (Parker-Consistent)

The filtered dataset used in this EDA follows the same quality-control philosophy as the Parker-style subduction preprocessing workflow implemented in `prepro.ipynb`. The objective is to keep physically valid, high-quality ground-motion records while preserving enough coverage across magnitudes, distances, event types, and regions.

3.1 Step-by-Step Filtering Workflow

Starting from the raw `NGAsub_flatfile.csv`, the preprocessing applies the following sequence:

1. **Basic amplitude and magnitude screening:** retain records with $M_w \geq 4$, $0 < \text{PGA}_g \leq 10$.
2. **Key input completeness:** remove records with missing values in `Earthquake_Magnitude`, `Rjb_km`, `Ztor_km`, `Hypocenter_Depth_km`, and `Vs30_Selected_for_Analysis_m_s`.
3. **Sentinel cleanup:** remove placeholder values (-999) used as missing-data sentinels in required predictors (notably `Rjb_km` and `Vs30`).
4. **Tectonic class selection:** keep only subduction records with `Intra_Inter_Flag` in $\{0, 1, 5\}$, corresponding to interface and intraslab classes.
5. **Depth plausibility constraints:** enforce physically consistent depth bounds for rupture and hypocenter metadata.
6. **Distance-domain filtering:** remove records outside the intended source-to-site distance applicability range.
7. **Near-surface sensor-depth control:** retain records with shallow sensor installation depth (e.g., ≤ 2 m) for site-response consistency.
8. **Single-source association:** remove multi-source or ambiguous source-attribution records.
9. **Signal quality screening:** remove records flagged for late P-wave arrival or other timing/quality problems.
10. **Station-condition filtering:** retain free-field records to avoid structural-response contamination.
11. **Source quality screening:** keep only high-quality source/event metadata entries.
12. **Usable period coverage:** retain records with adequate long-period usability (targeting at least 10 s period support).
13. **Event sufficiency rule:** keep events with at least 3 associated records to stabilize event-level effects.

3.2 Record Retention Trend

The notebook outputs show the expected Parker-style behavior: large early reductions from strict quality-control filters, followed by progressive refinement through reliability constraints. A representative retention chain printed by the preprocessing notebook is:

- Initial records: 71,340.
- After M_w /PGA checks: 68,375.

- After key-input missing-value removal: 68,375 (or 65,100 in the stricter variant that also removes sentinel-coded values earlier).
- After subduction event-type filtering: 45,719 (or 45,383).
- After depth and distance constraints: approximately 20,000 records.
- After sensor/source/signal-quality filters: approximately 17,000 to 18,000 records.
- After period-coverage and minimum-records-per-event constraints: final high-quality subset in the $\sim 2,000$ to $\sim 9,000$ range depending on strictness settings.

3.3 Why This Matters for the EDA

Using this preprocessing protocol improves the interpretability of all downstream figures and tables:

- Distribution plots are less distorted by invalid/sentinel values.
- Regional and tectonic comparisons are made on quality-controlled records.
- Residual decomposition is based on consistent metadata and usable spectral content.
- Reported predictor ranges better reflect practical model applicability bounds.

4 Plots and Images from the EDA Notebook

4.1 Magnitude Distributions by Event Type

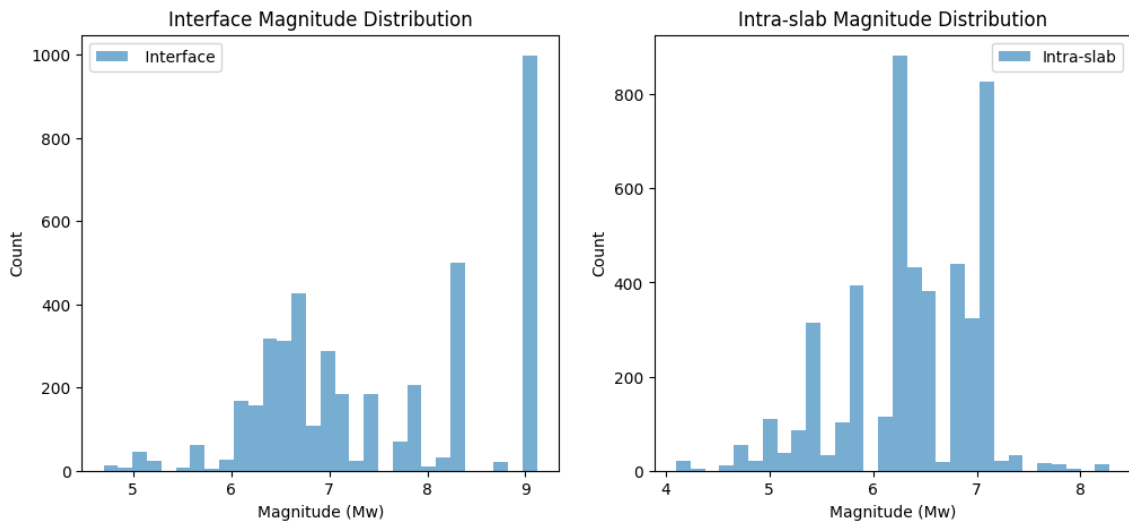


Figure 1: Side-by-side histograms of interface and intraslab magnitude distributions.

Inference:

- Both classes span moderate-to-large magnitudes, confirming that the dataset supports subduction scaling analysis across a broad M_w range.

- Sparse counts in the highest-magnitude tail imply higher epistemic uncertainty for extreme-event predictions.

4.2 Magnitude vs Distance

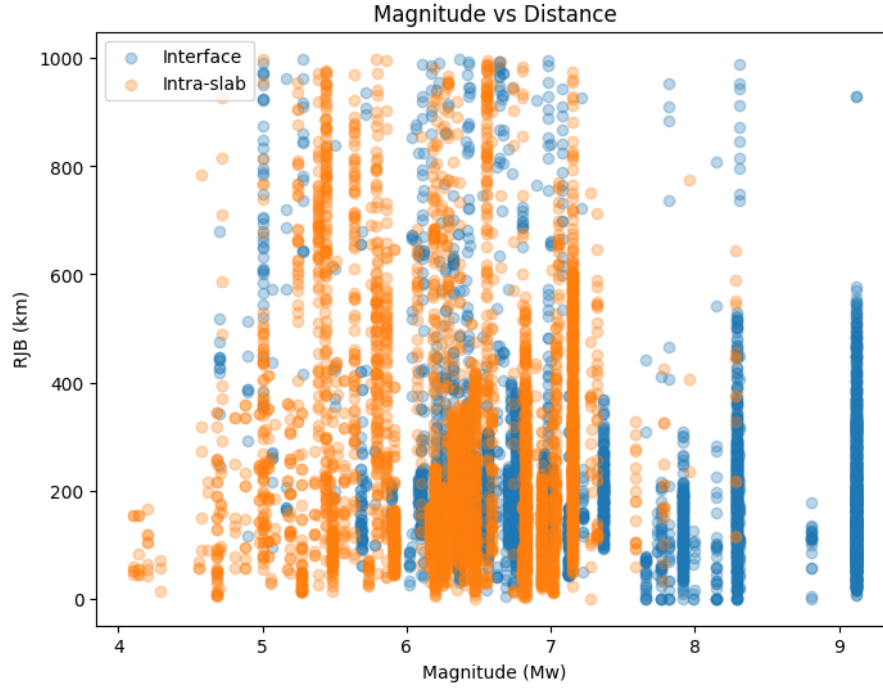


Figure 2: Scatter of magnitude (M_w) versus Joyner–Boore distance (R_{jb}).

Inference:

- The cloud shows broad distance coverage, favorable for separating source scaling from path attenuation.
- Data concentration is visibly stronger at shorter-to-moderate distances.

4.3 Region Distribution

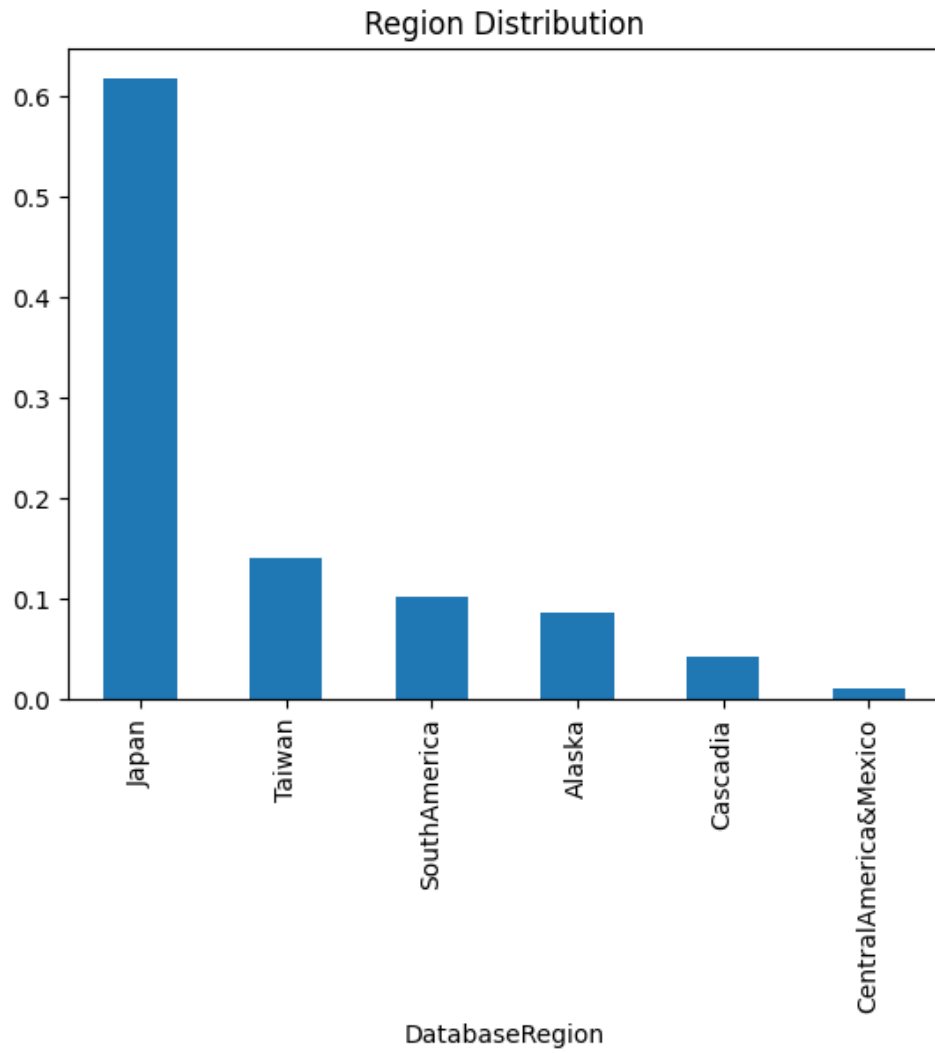


Figure 3: Normalized frequency of records by database region.

Inference:

- The regional bars are not uniform; regions with larger fractions will dominate global loss functions unless balanced.

4.4 Depth Distribution

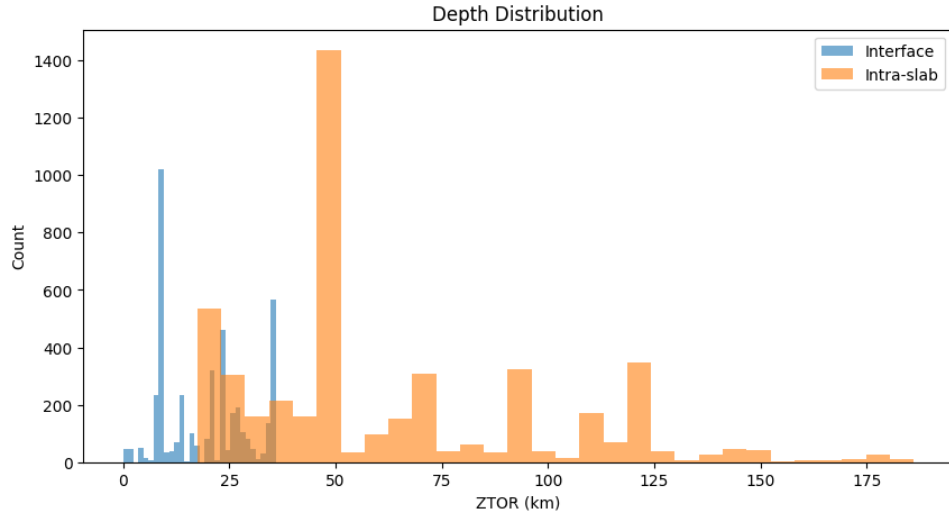


Figure 4: Histogram comparison of Z_{tor} for interface and intraslab events.

Inference:

- Intraslab events occupy deeper Z_{tor} bins than interface events, consistent with physical subduction geometry.

4.5 V_{S30} Distribution

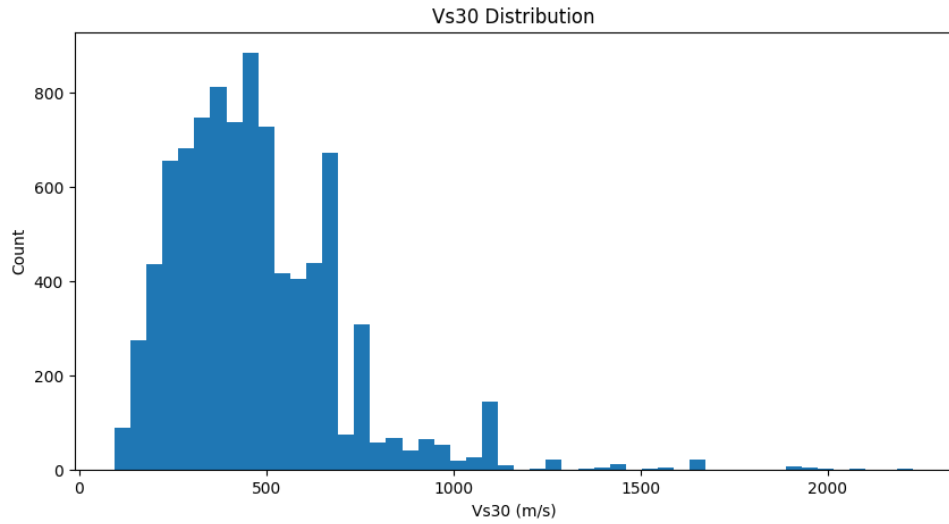


Figure 5: Distribution of V_{S30} values across all records.

Inference:

- The dataset includes a wide V_{S30} domain (soft to stiff sites), enabling analysis of site amplification trends.

4.6 Records, Events, and Stations by Region

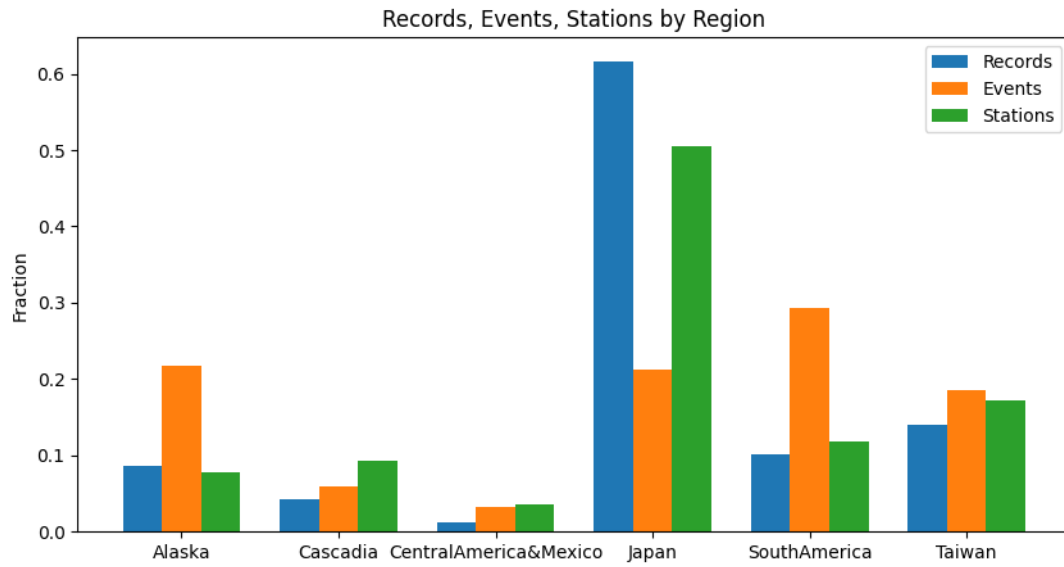


Figure 6: Regional fractions of records, unique events, and stations.

4.7 Tabular Summary Rendered as Figure

Region	Mw_min	Mw_max	RJB_min	RJB_max	ZTOR_min	ZTOR_max	Vs30_min	Vs30_max
Alaska (interface)	5.0	7.21	39.4975025	999.0898298	9.915	32.878	322.0	750.0
Alaska (intraslab)	5.0	7.96	28.1045623	998.5433826	34.548	154.585	210.0	750.0
Cascadia (interface)	4.7	4.9	114.8384898	678.7427604	12.279	17.908	376.0	665.0
Cascadia (intraslab)	4.58	6.8	3.5169528	988.1309973	18.0	52.17	140.0	2000.0
CentralAmerica&Mexico (interface)	6.32	7.99	0.0	747.2489218	6.0	29.81	225.0	586.0
CentralAmerica&Mexico (intraslab)	6.47	7.32	107.6496923	945.5241019	36.43	158.566	281.0	2100.0
Japan (interface)	5.99	9.12	0.0	928.831581	1.22	36.011	94.7	2229.8
Japan (intraslab)	5.48	8.28	0.0	644.8758255	17.5	120.3534	94.7	1909.1
SouthAmerica (interface)	6.1	8.81	0.0	995.0439642	0.0	33.973	198.0	1951.0
SouthAmerica (intraslab)	6.07	7.78	0.0	995.1056446	33.28	186.004	116.0	1639.0
Taiwan (interface)	5.68	7.12	42.4643494	678.8855344	3.96	22.105	159.0	1037.8827
Taiwan (intraslab)	4.1	7.02	4.3673402	654.2300751	24.13	177.603	121.4493	1037.8827

Figure 7: Range summary table rendered as an image in the notebook.

4.8 Regional Scatter by Event Type

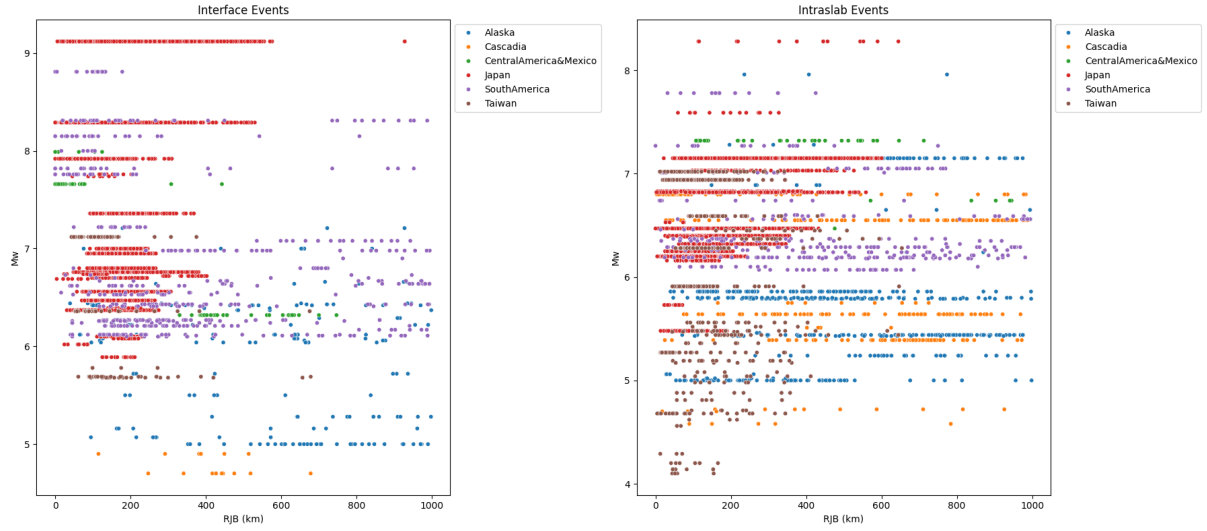


Figure 8: Interface and intraslab scatterplots of M_w versus R_{jb} , color-coded by region.

4.9 Regional Histograms of Key Predictors

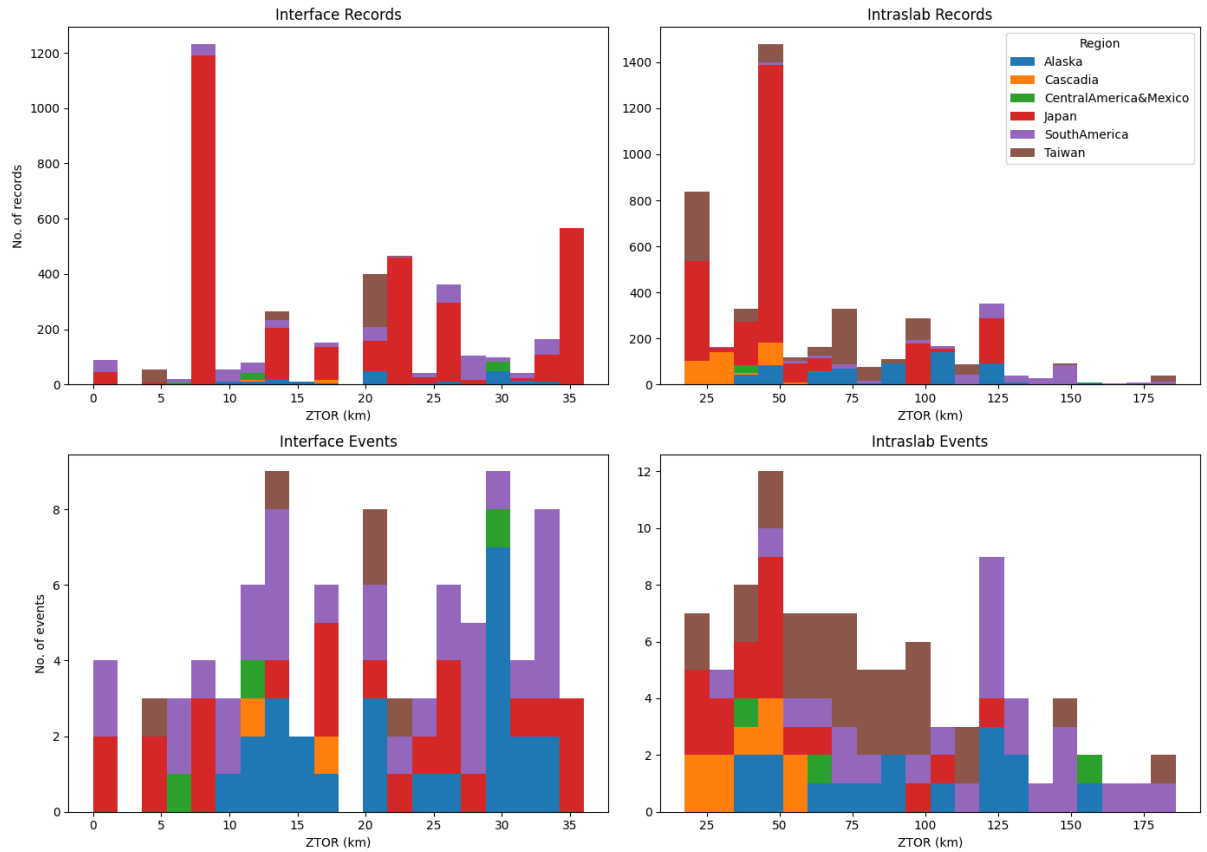


Figure 9: Multi-panel regional histograms for selected predictor variables.

4.10 Interface vs Intraslab Regional Histograms

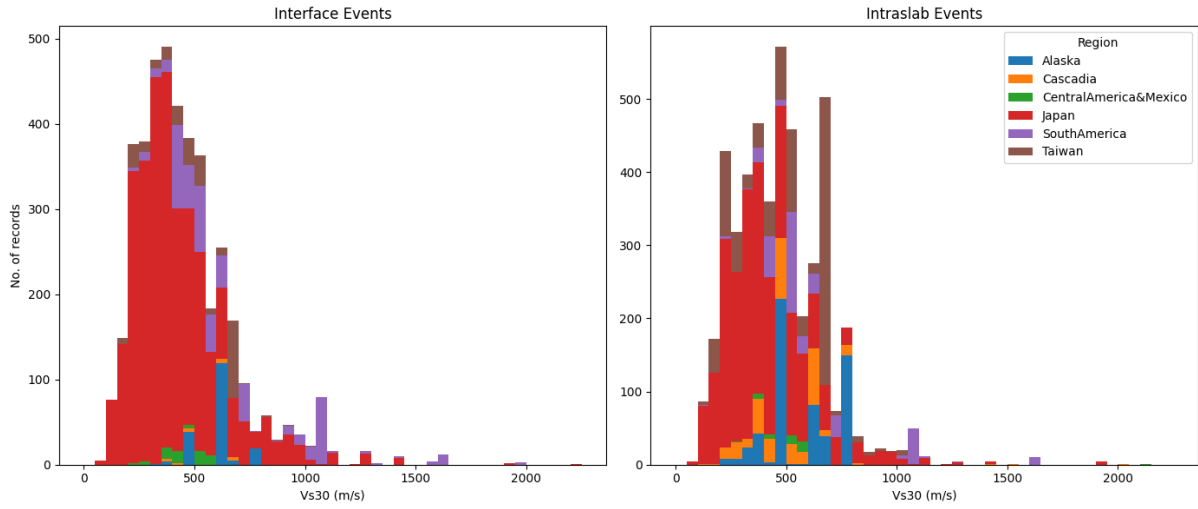


Figure 10: Regional histograms separated by event type for comparative coverage assessment.

5 Tables from the EDA

5.1 Table 1: Regional Predictor Ranges

Table 1: Summary of predictor ranges by region and tectonic class.

Region	Mw							
<i>min</i>	Mw							
<i>max</i>	RJB							
<i>min</i>	RJB							
<i>max</i>	ZTOR							
<i>min</i>	ZTOR							
<i>max</i>	Vs30							
<i>min</i>	Vs30							
<i>max</i>								
Alaska (interface)	5.0	7.21	39.4975025	999.0898298	9.915	32.878	322.0	750.0
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Japan (interface)	5.89	9.12	0.0	928.831581	1.22	36.011	94.7	2229.8
Japan (intraslab)	5.48	8.28	0.0	644.8758255	17.5	120.3534	94.7	1909.1
SouthAmerica (interface)	6.1	8.81	0.0	995.0439642	0.0	33.973	198.0	1951.0
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Taiwan (interface)	5.68	7.12	42.4643494	678.8855344	3.96	22.105	159.0	1037.8827
Taiwan (intraslab)	4.1	7.02	4.3673402	654.2300751	24.13	177.603	121.4493	1037.8827

This table defines practical modeling bounds by region and event class for M_w , R_{jb} , Z_{tor} , and V_{S30} .

5.2 Table 2: Residual Means and Standard Deviations

Table 2: Residual mean and standard deviation decomposition across periods.

Period	Inter-event mean	Inter-region mean	Intra-event mean	Inter-event std	Inter-region std	Intra-event std	Sigma
0.01	-0.0002	0.0	0.0	0.3767	0.0	0.7146	0.8078
0.02	-0.0003	-0.0	-0.0	0.3798	0.0	0.7169	0.8113
0.03	-0.0013	-0.0	-0.0	0.3878	0.0	0.726	0.823
0.04	-0.0012	0.0	0.0	0.3977	0.0	0.7412	0.8412
0.05	-0.0023	-0.0	-0.0	0.4077	0.0	0.7605	0.8629
0.1	-0.0	-0.0	-0.0	0.4384	0.0	0.8327	0.941
0.2	0.0037	0.0	-0.0	0.4114	0.0	0.8095	0.9081
0.3	0.0055	0.0	0.0	0.3924	0.0	0.7581	0.8536
0.5	0.0053	0.0	0.0	0.3678	0.0	0.6996	0.7904
0.75	0.0023	0.0	-0.0	0.3636	0.0	0.6805	0.7716
1.0	0.0013	0.0	0.0	0.3645	0.0	0.676	0.768
2.0	-0.004	-0.0	-0.0	0.376	0.0	0.6758	0.7734
5.0	-0.0044	-0.0	0.0	0.3553	0.0	0.5998	0.6972
10.0	0.0021	0.0	-0.0	0.3398	0.0	0.4999	0.6044
0.0	0.0001	-0.0	-0.0	0.3762	0.0	0.7133	0.8064
-1.0	0.0013	0.0	0.0	0.3181	0.0	0.5548	0.6396
-2.0	0.0016	-0.0	-0.0	0.3165	0.0	0.5335	0.6203

This table decomposes residual variability into inter-event, inter-region, and intra-event terms.

6 Interpretation Highlights

- The EDA confirms broad magnitude-distance coverage.
- Interface and intraslab classes show distinct depth behavior, validating separate treatment.
- The standard-deviation table indicates period-dependent scatter, suggesting variable predictive uncertainty.